SFT

SFT + GRPO

**SFT only**
no tools, 1,847 single-turn traces

**SFT + GRPO**
no tools, same recipe, single-turn

no tools



one pass → answer

H2: RL helps?



one pass → answer

H1: tools help?

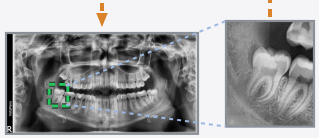
H1: tools help?

**SFT + tools**
1,847 with-tools traces

**VLM-DENTAL**
SFT + GRPO + tools, live tool rollouts

with tools

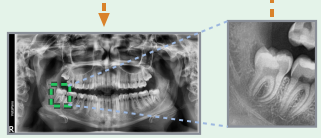
look again



answer

H2: RL helps?

look again



answer

F1 and invented findings of all four: pending the first hardware run

**Zero-shot references**
exact F1

best frontier VLM **0.222**

Qwen3.5-9B **0.112**

majority floor **0.038**

**the four conditions: pending**

**H3: supervised specialist**
HierarchicalDet on DENTEX

**AP 0.372**
**AP50 0.601 · AR 0.653**

**a fair test,**
**not a sure win**

**Evaluation protocol, frozen**

**test**
DENTEX official validation, used once

46 pathological | 4

50 images, 182 target teeth

**dev**
about 70 training images
one fixed-seed draw
every choice is made here

**metrics**
pooled F1, set-matched, tooth-only
invented findings on healthy images
no-answer rate, ECE

**uncertainty** bootstrap over images, paired

**probes**
on the trained checkpoint,
nothing retrained:
tool removal
grounding perturbation
group size K in {1,2,4,8,16}